Market Selection with Midpoint Matching: A Strategic Agent-Based Analysis

Gabriel Smithline
Computer Science and Engineering
University of Michigan
Ann Arbor, MI, USA
gsmithl@umich.edu

Anri Gu
Computer Science
University of Chicago
Chicago, IL, USA
anrigu@uchicago.edu

Michael Wellman
Computer Science and Engineering
University of Michigan
Ann Arbor, MI, USA
wellman@umich.edu

Abstract

We study midpoint matching through Nasdaq's Midpoint Extended-Life Order (M-ELO) mechanism, which offers non-displayed midpoint execution subject to a mandatory holding period to balance liquidity and adverse-selection concerns. We conduct an empirical game-theoretic analysis using an agent-based simulation in PyMarketSim that models both M-ELO and a traditional lit order book, simulating traders' venue choices across a range of holding periods and enumerating pure and mixed Nash equilibria for each setting . We analyze welfare outcomes, strategic stability via deviation graphs, and basins of attraction across these equilibria . Our findings reveal that shorter holding periods can insufficiently deter higher frequency small lot trading, while excessively long delays erode the midpoint premium for large traders. This highlights the fundamental trade-off in designing holding periods that effectively screen predatory trading without undermining value for intended users.

Keywords

Dark Pools, M-ELO, Empirical Game-Theoretic Analysis

ACM Reference Format:
Gabriel Smithline, Anri Gu, and Michael Wellman. 2025. Market Selection with Midpoint Matching: A Strategic Agent-Based Analysis. In *6th ACM International Conference on AI in Finance (ICAIF '25), November 15–18, 2025, Singapore, Singapore.* ACM, New York, NY, USA, 9 pages. https://doi.org/10.1145/3768292.3770419

1 Introduction

A ***dark pool*** market mechanism admits and matches orders that are not displayed publicly, at execution prices typically determined by prices on "***lit***" markets where orders are displayed. Such mechanisms are designed to appeal to relatively large, slow, and/or uninformed traders, who seek to avoid price impact and adverse selection. To achieve the desired effect, the mechanism must somehow ensure sufficient activity in the lit market for effective price discovery, and deter the fast and informed traders from preying on orders in the dark pool.

An example mechanism supporting dark trading is Nasdaq's ***Midpoint Extended-Life Order*** (M-ELO), introduced by the exchange in 2018. M-ELO orders are undisplayed and execute at the midpoint of the lit market, as defined by the ***National Best Bid and Offer*** (NBBO): the effective spread aggregated over public exchanges. Figure 1 illustrates the connection between lit and dark markets. To sort the intended and unintended uses, M-ELO orders are subject to a specified ***holding period*** before they become eligible for execution. This period was initially set at 500 milliseconds, and in 2020 was reduced to 10 milliseconds to promote broader participation. In 2023, Nasdaq deployed an adaptive version, ***Dynamic M-ELO*** (D-MELO) [14], where holding periods are continually adjusted using a reinforcement-learned policy responsive to real-time market conditions.

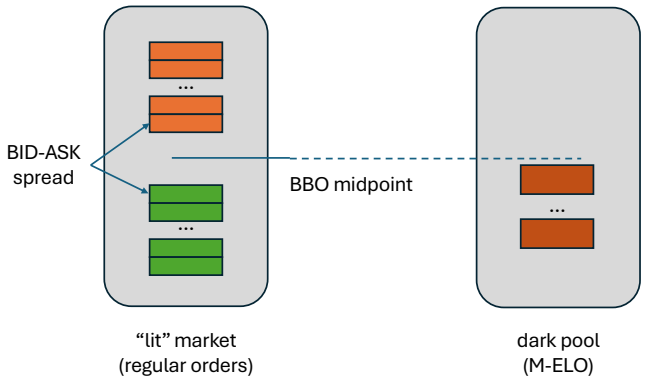


Figure 1: Lit and dark markets connected through a midpoint pricing mechanism. The lit market operates as a continuous limit-order book, here shown with outstanding sell orders in orange and buys in green. The dark pool may have larger sell orders priced more aggressively, but when a matching buy order comes in they will trade at the midpoint of the lit market.

Since its inception, M-ELO has gained substantial adoption, reflecting its popularity among institutional investors seeking reduced adverse selection risks and better execution quality [20]. At the end of 2023, Nasdaq reported significant growth in M-ELO volume, indicating its increasing prominence in the equity trading landscape [21]. Early empirical analyses indicate that M-ELO trading correlates with reduced incidence of mini-flash crashes, suggesting enhanced market stability [11, 16]. Given this robust adoption and evolving design, we take the Nasdaq M-ELO mechanism as the subject of our study of the use of midpoint matching. We design and implement a detailed simulation of M-ELO and lit-order trading in PyMarketSim [17], capturing holding period mechanics,

This work is licensed under a Creative Commons Attribution 4.0 International License.
ICAIF '25, Singapore, Singapore
© 2025 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2220-2/25/11
https://doi.org/10.1145/3768292.3770419

order processing queues, and heterogeneous trader strategies. In addition to modeling the M-ELO trading environment, we make the following contributions:

- We systematically apply empirical game-theoretic analysis to compute role symmetric Nash equilibria and measure welfare outcomes across holding periods from 0 to 380 time steps.
- We characterize the balance between deterring predatory trading and preserving midpoint-trading benefits for large institutional participants by comparing welfare outcomes across identified equilibria.
- We visualize how trader incentives shift as holding periods increase via deviation graphs and identify basins of attraction, illuminating the critical mass required for M-ELO adoption.

2 Background and Related Work

2.1 Dark Pools and Midpoint Matching

Dark pools are *non-displayed* trading venues that match buy and sell orders away from the visible (lit) limit-order book. The matched orders are priced most commonly at the midpoint of the associated lit market, but alternatives exist, including last sale prices, price-improvement pegging, or negotiated prices. Dark pools can reduce trading costs for those who need to trade at volume in short periods. A drawback of dark pools is that they may undermine price discovery, to the extent that relevant market information is hidden. Buti et al. [5] formalize this tradeoff, highlighting how the relative composition of informed versus uninformed order flow determines the net welfare impact of dark trading.

Empirical evidence confirms that dark-pool activity impacts market quality in a complex, nonlinear manner. Analyzing U.S. market data from 2016 to 2022 uncovers a U-shaped relationship between dark trading share and market quality indicators such as spreads and displayed depth [6]. Market quality initially improves with increased dark trading but deteriorates beyond approximately 15% market share. Cross-exchange studies further corroborate these findings: extensive hidden volume typically widens spreads and reduces depth [7], while decreased permanent price impacts suggest diminished informational efficiency as informed traders migrate into dark venues [31].

Given the complexity of modeling order-routing latency and multi-venue dynamics analytically, many researchers adopt agent-based modeling (ABM) and related financial simulation methods [2] to study dark pools. For instance, Mo et al. [18] developed an ABM featuring zero-intelligence and strategic liquidity traders to examine issues in midpoint execution, finding improved outcomes for large, slow traders yet identifying detrimental effects on price discovery. More recent simulation research incorporates high-frequency traders (HFTs) and latency arbitrage: HFT profits derived from exploiting stale dark pool quotes are quantified [1], and increased informational externalities as informed order flow shifts off-book are demonstrated [30].

Midpoint matching mechanisms have been operated by several exchanges. The ISE Stock Exchange introduced its Midpoint Match in 2006 [8]. The New York Stock Exchange's MatchPoint, introduced in 2009, likewise operates as a dark pool matching midpoint-priced orders, executed at scheduled intervals [22]. Brolley [4] reports that midpoint execution mechanisms were widely adopted among dark pools. Studies by Bayona et al. [3] and Buti et al. [5] further examine interactions between dark and lit markets, identifying strategic venue-selection behaviors and market-quality impacts.

Nasdaq's M-ELO mechanism operates as a dark pool with a holding period for order entry. The holding timer begins immediately if the order's limit price is at or better than the NBBO midpoint upon submission; otherwise, it starts when the midpoint moves within the order's limit. Importantly, subsequent changes to the NBBO do not reset the timer. After the delay elapses, M-ELO queue in strict time priority and may trade only against other active M-ELO orders at the prevailing midpoint price [19].

2.2 Market Selection Games

Market selection games model strategic interactions wherein traders have a choice among multiple competing venues. Analyzing strategic choices in market selection can shed insight on whether proposed mechanisms will be voluntarily adopted. For example, Wah et al. [26] studied strategic adoption of periodic call markets versus continuous markets, among fast and slow traders. They identified a predator-prey dynamic, where the slow traders generally preferred the protection of the call markets, and the fast traders preferred to go wherever the slow traders were.

Shi et al. [25] present an analytical framework for venue choices in competing double-auction environments. They find potential instabilities and path dependencies in venue coexistence. In decentralized market selection games studied by Ishibuchi et al. [13], efficient liquidity distribution across multiple venues emerges spontaneously from individual adaptive behaviors without centralized coordination.

2.3 Empirical Game-Theoretic Analysis (EGTA)

Empirical game-theoretic analysis (EGTA) is a computational approach to studying strategic interactions in complex games [27]. EGTA constructs a game model based on simulations of agent interactions. This method is particularly salient for environments that are too complex for analytical game modeling, such as strategic decision-making in financial markets. For example, EGTA was employed in some early studies of heuristic strategies for trading on continuous-time limit order books [15, 23].

A straightforward way to apply EGTA is to collect simulation data for all strategy profiles. With N players and S strategies, there are N^S profiles, and even if the game is symmetric the number of profiles is exponential in $\min(N, S)$. Financial market scenarios commonly involve many agents, thus we employ a technique called ***deviation-preserving reduction*** (DPR) [28] designed for scaling EGTA in players. The key idea of player reduction is to approximate a many-player symmetric game by one with a small number of players. Specifically, in the reduced p-player game ($p \ll N$), the payoff to playing strategy s when the other $p-1$ players each play $(s_1, \ldots, s_{p-1})$ is given by the payoff in the full N-player game when playing s against the $N-1$-player profile where $\frac{N-1}{p-1}$ other players play $s_1, \ldots, s_{p-1}$, respectively.

DPR models the p players as controlling one full-game player, while the other players are aggregated. This has the property of ensuring that a one-player deviation is beneficial in the reduced

game iff it is beneficial in the full game. Thus in particular, role-symmetric pure-strategy Nash equilibria (PSNE) in the reduced game are also role-symmetric PSNE of the full game.

3 Agent-Based Financial Market

3.1 Market Model

Our market model extends those used in previous studies [12, 24]. It employs a continuous limit order book (LOB) mechanism over a single security. Time progresses in integer time steps, $t \in [1, T]$. Agents in the model submit buy (sell) limit orders, specifying the maximum (minimum) price they are willing to pay (accept) and the number of units they wish to trade.

Fundamental Value Process. The dynamic intrinsic value of the security is determined by a mean-reverting stochastic process:

$$r_t = \max\{0, \kappa\bar{r} + (1-\kappa)r_{t-1} + u_t\}; r_0 = \bar{r},$$

where r_t is the *fundamental value* at time t; the mean-reversion rate $\kappa \in [0, 1]$ captures the tendency of the fundamental to revert to its mean $\bar{r}$; $u_t \sim \mathcal{N}(0, \sigma_s^2)$ represents a random shock applied to the fundamental at each step, σ_s^2 being the shock variance. The *realized* value of the security at the end of the finite horizon T is given by r_T.

Given an observation of the current fundamental, agents can compute $\tilde{r}_t$, an estimate of r_T:

$$\tilde{r}_t = (1 - (1-\kappa)^{T-t})\bar{r} + (1-\kappa)^{T-t} r_t. \tag{1}$$

Limit Order Book. The market tracks outstanding orders using a *limit order book*. The book has two sides organized as priority queues: the buy (sell) side prioritizes orders by higher (lower) limit prices and earlier arrival times, further ties being broken uniformly at random. When orders are transacted or withdrawn, the order book is updated instantly by appropriate order removals or quantity adjustments.

Let BID_t (ASK_t) denote the highest buy offer (lowest sell) price at time t, defining the ***spread*** at that time. For lit markets, these are displayed as price quotes. The ***midpoint***

$$BBO_t = \frac{BID_t + ASK_t}{2}$$

is the price at the middle of the spread.

Arrival Process. Agents (re-)arrive to the market following a Poisson process with an agent-specific arrival rate λ_a. Upon each arrival, an agent first withdraws its previous outstanding orders from the order book and then submits new limit order(s) based on its observations and its trading strategy. Because order prices are based on information known on arrival, standing orders from agents who arrive relatively more frequently on average reflect more accurate estimates of the fundamental value.

3.2 Trading Strategies

We consider two categories of market participants:

(1) ***Background traders*** represent the standard trader with relatively little concern about adverse selection and price impact. These traders arrive to the market frequently and trade a single unit at a time.

(2) ***MOBI*** (M-ELO Open Book Interaction) ***traders*** have larger blocks to trade and arrive less frequently, so can be potentially exploited by the background traders. MOBI agents represent the target population for dark pools—that is, who M-ELO was designed for.

Both trader categories have trading objectives defined as a combination of common (fundamental) and private values. They also both employ a variant of the zero-intelligence (ZI) heuristic trading strategy for pricing orders. As part of their strategies, they must also choose which market mechanism to use: lit or M-ELO.

Background Traders. The net position q (positive for long, negative for short) of a background trader is an integer representing the number of units of the security held and is constrained to be at most $q_{\max}$ in magnitude. Each background trader i has an individual *private value* expressed as a vector Θ_i of length $2q_{\max}$:

$$\Theta_i = [\theta_i^{-q_{\max}+1}, \ldots, \theta_i^{-1}, \theta_i^0, \theta_i^1, \ldots, \theta_i^{q_{\max}}],$$

where θ_i^q is the incremental private benefit forgone by *selling* one unit (alternatively, θ_i^{q+1} is the marginal private gain from *buying* one additional unit) given the current position q. We construct Θ_i by generating $2q_{\max}$ samples independently from the distribution $\mathcal{N}(0, \sigma_{PV}^2)$ and sorting them in descending order, thus ensuring diminishing marginal returns. That is, $\theta^{q'} \leq \theta^q$ for all $q' \geq q$.

Moreover, upon every arrival, a background trader is designated to be a buyer or seller with equal probability and places a new order of unit quantity.

ZI Strategy. The ZI trading strategy was originally introduced by Gode and Sunder [10] as a way of demonstrating that strategic sophistication was not necessary to approximate competitive equilibrium outcomes. Subsequent studies have found that ZI agents can generate market behavior with other realistic patterns as well [9]. Wright and Wellman [29] showed that equilibrated ZI agents leave relatively little room for improvement by strategies derived via tabular reinforcement learning.

Upon each arrival, our ZI agent observes the current fundamental r_t and uses it to compute an estimate $\tilde{r}_t$ of the final fundamental via Equation (1). Its limit price is a combination of this estimate, its private value, and a random offset $o \sim \mathcal{U}(R_{\min}, R_{\max})$ where $R_{\min}$ and $R_{\max}$ are lower and upper bounds on the ZI agent's demanded surplus. For agent i with current position q, this price is given by

$$p_i = \tilde{r}_t + \begin{cases} \theta_i^{q+1} - o & \text{if buying,} \\ \theta_i^q + o & \text{if selling.} \end{cases} \tag{2}$$

MOBI Traders. Compared to background traders, MOBI traders place orders in higher quantities, and they arrive to the market at a slower rate. The private valuation for MOBI traders differs qualitatively from background traders. MOBI private values are constant per-unit (rather than diminishing marginal value), and independent across different arrivals. They draw a new private value from a distribution $\Phi = \mathcal{N}(0, \sigma_{PV}^2)$ at each arrival, which applies only to trades they may make before their next arrival.

Our MOBI agent likewise adopts the ZI strategy for pricing orders. Specifically, a MOBI i arriving at time t sets the order price per Equation (2), ignoring superscripts on the θ_i. When trading in the M-ELO market, MOBIs set $o = 0$. There is less reason to

shade their M-ELO offer, as midpoint pricing means that their own order does not set the price. The dark pool also makes them less vulnerable to adverse selection. When trading in the lit market, $o \sim \mathcal{U}(R_{\min}, R_{\max})$.

3.3 Payoffs

At the end of each simulation run, player payoffs are calculated as the realized surplus from their trading over the period. Surplus for each background agent i is computed as the sum of their net cash ($cash_i$) and the valuation $final_val_i$ of their final net position q_i^T:

$$surplus_i = final_val_i + cash_i.$$

The final valuation in turn is determined by the final fundamental r_T, and the agent-specific cumulative private value of the position:

$$final_val_i = r_T \cdot q_i^T + \begin{cases} \sum_{k=1}^{k=q_i^T} \theta_i^k & \text{if } q_i^T > 0, \\ -\sum_{k=q_i^T+1}^{k=0} \theta_i^k & \text{if } q_i^T < 0. \end{cases}$$

Since MOBI traders draw private values for each arrival independently, their cumulative private values depend on which arrivals their trades are associated with, and so cannot be calculated based on final position. Therefore we record private values associated with MOBI agent trades throughout the simulation, and sum them to compute the total private value in the final state.

4 M-ELO Market Structure and Implementation

We implement M-ELO as a new market mechanism within PyMarketSim. The mechanism manages M-ELO orders in three queues, depicted in the flow diagram of Figure 2:

(1) an *eligibility queue* for orders waiting to cross the market midpoint,
(2) a *holding queue* for orders undergoing the mandatory holding period, and
(3) an *active queue* for execution-ready orders.

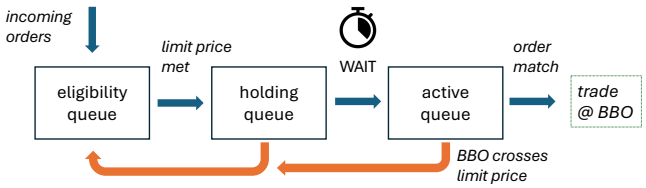


Figure 2: M-ELO order processing. Incoming orders enter the eligibility queue, where they remain until their limit price is marketable against the current lit market midpoint, *BBO*. Once marketable, an order moves into the holding queue for the mandatory delay. After waiting for the holding period the order becomes active and can execute at the prevailing midpoint. If, at any time, the midpoint moves to a price worse than the order's limit price, the order is returned to the eligibility queue.

4.1 Order Processing Workflow

When a market participant submits a M-ELO order, the mechanism routes it to either the eligibility queue or the holding queue based on its relation to the current lit market midpoint, *BBO*. Consider a buy M-ELO placed at limit price p: if the order is executable at the lit market midpoint (i.e., $BBO \leq p$), the order has *crossed* the *BBO* and progresses directly to the holding queue to fulfill the mandatory holding period. Conversely, if $BBO > p$, the order enters the eligibility queue to wait for a change in *BBO* that would make it executable. The inequalities are reversed for a sell M-ELO.

Upon completion of the holding period, the order transitions from the holding queue to the active queue, where it becomes eligible for midpoint execution against compatible M-ELO orders. However, in the event of a *BBO* change that crosses against the order, the order is returned to the eligibility queue to await a favorable change in *BBO*.[1]

4.2 Queue Implementation Specifications

Eligibility Queue. The eligibility queue is implemented as a price-prioritized heap, `heap[Order]`. Buy orders are organized in descending price order, while sell orders follow ascending price priority. At each time t when BBO_t is updated, the system evaluates whether the top order in each of the buy and sell heaps crosses BBO_t. Upon crossing, the order transfers to the holding queue, with the timestamp t recorded as the start of its holding period. The evaluation proceeds iteratively through the heap until reaching an order that does not cross BBO_t, at which point it terminates.

Holding Queue. The holding queue follows a first-in-first-out (FIFO) discipline, `queue[(Order, enter_time)]`, where each element consists of an order paired with its entry timestep. The mechanism evaluates this queue at relevant time steps to identify orders that have completed their prescribed holding period. Upon fulfillment of their temporal holding requirement, orders are passed onto the active queue.

Active Queue. At any moment, the active queue will contain exclusively (INTERRUPT) The active queue operates as a FIFO structure as well: `queue[Order]`. Since M-ELO execution priority is determined by temporal precedence rather than price, FIFO suffices for sequencing execution-ready orders. Recall that an order is sent back to the eligibility queue if the midpoint exceeds its reserve price at execution time.

5 Experiments

5.1 Market and Analysis Setup

Our experiments are run using PyMarketSim, with a standard continuous LOB as the lit market, in conjunction with the M-ELO mechanism. Our environment contains 68 traders, divided into two roles: 28 MOBI agents and 40 background traders. Market environment parameters are fixed across runs: the fundamental value has mean $\bar{r} = 1 \times 10^6$, shock variance $\sigma_s^2 = 1 \times 10^6$, and

[1]This differs from Nasdaq's implementation [19]. In the actual mechanism, M-ELO orders once active remain so even if the midpoint crosses their limit price. Traders run the risk of getting a worse execution than their limit; it is up to them to monitor their orders and take action when necessary. Our approach is more conservative, guaranteeing that the original limits are respected.

mean-reversion rate $\kappa = 1 \times 10^{-3}$. The horizon is $T = 1 \times 10^4$ time steps. Trader arrivals follow Poisson processes with rates $\lambda_{\text{Bkg}} = 6{\times}10^{-3}$ (one every 167 steps on average) for background traders and $\lambda_{\text{MOBI}} = 1 \times 10^{-3}$ (one every 1000 steps) for MOBI agents. The key strategic choice for each player role is which market to participate in: lit or M-ELO. Given this choice, the agent submits all orders to the chosen market. ZI strategy parameters are $[R_{\min}, R_{\max}] = [250, 500]$. MOBI agents shade only in the lit venue (i.e., their M-ELO orders are priced at valuation), whereas background traders shade with the same surplus-offset range in both markets. Background traders trade with an order size of 1, and MOBI traders trade with an order size of 5. Each background trader draws $2q_{\max}$ marginal private values, with $q_{\max} = 10$, sampled from $\mathcal{N}(0, 5 \times 10^6)$. Each MOBI trader draws a new private value upon arrival sampled from $\mathcal{N}(0, 5 \times 10^6)$.

We systematically vary the holding period at intervals of 20, $HP \in \{0, 20, \ldots, 380\}$. To mitigate stochastic variability from the fundamental path, Poisson arrivals, and private valuations, each strategy profile is simulated $R = 50{,}000$ times; recorded payoffs are averages over these Monte-Carlo draws.

For each *HP* setting, we analyze an empirical game to identify the equilibrium market choices. Even with only two strategies per role (i.e., the market choice), simulating all profiles over 68 players would be imposing. We therefore employ DPR to estimate approximate 8-player games (4 MOBI and 4 background). We collect simulation data for all strategy profiles in the full (68-player game) that are relevant for estimating payoffs in the $(4+4)$-player DPR game. The process is as follows:

(1) DPR profiles. Enumerate all tuples (n_1, n_2), $n_i \in \{0, 1, 2, 3, 4\}$. n_1 represents the number of MOBI players choosing M-ELO, and n_2 the number of background traders choosing M-ELO. We write the corresponding DPR profile as $(n_1, 4-n_1; n_2, 4-n_2)$.
(2) Required full-game profiles: For each DPR profile $(n_1, 4-n_1; n_2, 4-n_2)$, collect the following full-game profiles to estimate the payoffs to each role/strategy.
   - MOBI playing M-ELO: If $n_1 = 0$, n/a; otherwise $(1+9(n_1-1), 10n_2)$. That is, the full-game profile with $1+9(n_1-1)$ MOBIs choosing M-ELO and $10n_2$ Backgrounders choosing M-ELO.[2]
   - MOBI playing lit: If $n_1 = 4$, n/a; otherwise $(1+9(3-n_1), 10n_2)$.
   - Background playing M-ELO: If $n_2 = 0$, n/a; otherwise $(7n_1, 1+13(n_2-1))$.
   - Background playing lit: If $n_2 = 4$, n/a; otherwise $(7n_1, 1+13(3-n_2))$.
(3) Simulation and payoff estimation: Simulate the required set of full-game profiles specified above, each for R replications. Record the mean payoff for each role and strategy.
(4) Reduced-game construction: Assemble the two-role $(4+4)$-player DPR game using the full-game payoffs obtained above.
(5) Equilibrium computation: Identify all PSNE of the DPR game (25 profiles, so exhaustive testing is feasible). Search for role-symmetric mixed NE using replicator dynamics from a variety of initial points.

5.2 Equilibrium and Welfare Analysis

Before even analyzing any empirical games, we know that there are two equilibria for every value of *HP*, namely the pure profiles where all players go to M-ELO, or all go to the lit market. The reason is simple: if all other players are in one market, then it is impossible to trade in the other. Actually, the situation for all-M-ELO is even worse. With nobody in the lit market, there is no midpoint to enable trading, yet it is still an equilibrium because exactly one player in lit does not help. Thus, we consider the all-M-ELO equilibrium degenerate.

Figures 3 and 4 present the welfare (sum of surplus over players) of these profiles as well as other equilibrium configurations, as a function of *HP*. The blue points correspond to the degenerate all-M-ELO equilibrium, which of course has zero payoff for everyone. The all-lit equilibrium is in orange, and exhibits relatively high payoffs. These are invariant to *HP*, which affects only what happens in the M-ELO market.

A more interesting effect of *HP* is revealed in the profile (shown in green) where MOBIs all trade in M-ELO and backgrounders stay in the lit market. As *HP* increases, payoffs to MOBI naturally decline, as their orders have to spend more of their time in the holding queue. Background traders experience no welfare impact from *HP* since they stay in the lit market. Note that background traders benefit substantially from the presence of MOBIs in the lit market, as illustrated by the jump in welfare between the green and orange curves. This results from the extra surplus background traders are able to extract by trading with the lower-frequency MOBIs. For MOBIs, trading in M-ELO with backgrounders in the lit market (green profile) yields the highest welfare for most of the *HP* range, falling below the all-lit equilibrium (orange) only at the high end of the range ($HP \geq 320$).

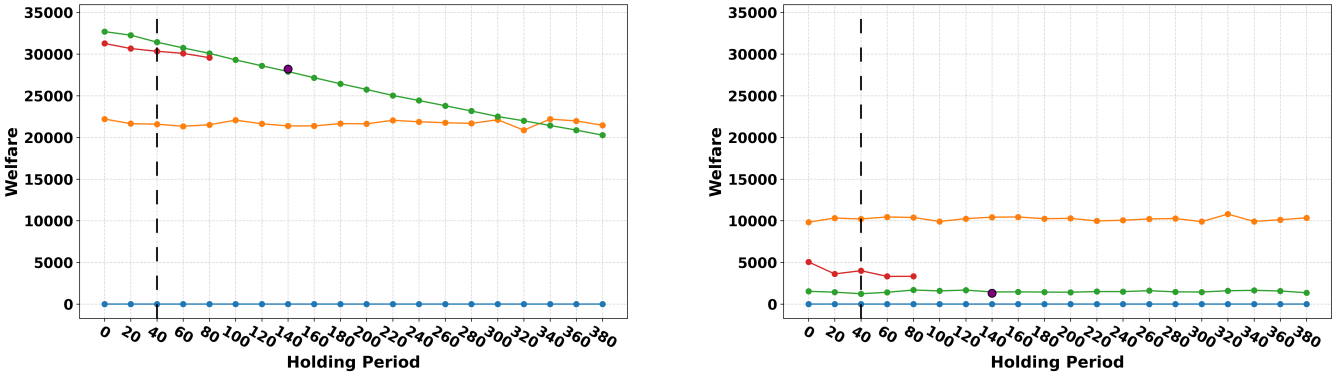
The welfare plots (Figures 3 and 4) also illustrate the distinct equilibrium regimes. As noted, the all-M-ELO and all-lit equilibria (blue and orange) always exist. For $HP \leq 80$ the DPR game exhibits a PSNE (red) at (4,0;3,1): every MOBI trades in M-ELO, while background traders mostly follow suit, leaving only one of the four in the lit market.[3] For $60 \leq HP \leq 120$ and $160 \leq HP$ the profile (green) with all MOBIs in M-ELO and all backgrounders in lit is in equilibrium.[4] Finally, at $HP = 140$ we find (in lieu of the green PSNE) a similar mixed equilibrium (purple) where MOBIs trade exclusively in M-ELO, while background traders trade in the lit market with probability 0.91.

As the pattern found shows, for all holding periods there is a non-degenerate equilibrium where MOBIs trade in M-ELO. At low *HP*, the M-ELO equilibrium (red) also includes many background traders there. As *HP* increases, this equilibrium goes away and is replaced with the green equilibrium where MOBIs have the M-ELO market to themselves. Note that compared to the all-lit equilibrium

[2] To explain this calculation, the MOBI playing M-ELO in the DPR game considers itself as one playing M-ELO in the full game, with the other 27 MOBIs evenly divided among the other 3 DPR MOBI players. Thus, $27/3 = 9$ for each of the $n_1 - 1$ other DPR MOBI players also play M-ELO in the full game. From the MOBI perspective, the 40 full-game Background players are evenly divided according to how the 4 DPR Backgrounders are assigned.

[3] This is not a PSNE in the full 68-player game.

[4] As this is a symmetric pure profile, being a PSNE in the DPR game entails that it is also a PSNE of the corresponding full game.



(a) MOBI surplus

(b) Background-trader surplus

Figure 3: Total surplus obtained in equilibrium profiles, as a function of *HP*, for (a) MOBI traders and (b) background traders. Blue: degenerate equilibrium in which both MOBIs and background traders all trade in M-ELO. Orange: both MOBIs and background traders all trade in the lit market. Green: all MOBIs choose M-ELO and all background traders choose the lit market. The black dashed line delimits the region to the right ($HP \geq 60$) where the green profile is in equilibrium (except for $HP = 140$). Red: profile where all MOBIs trade in M-ELO along with 3/4 of the background traders. This is based on the DPR PSNE found for $0 \leq HP \leq 80$. Purple: mixed equilibrium where MOBIs trade in M-ELO and background traders trade with probability 0.09 in M-ELO ($HP = 140$ only).

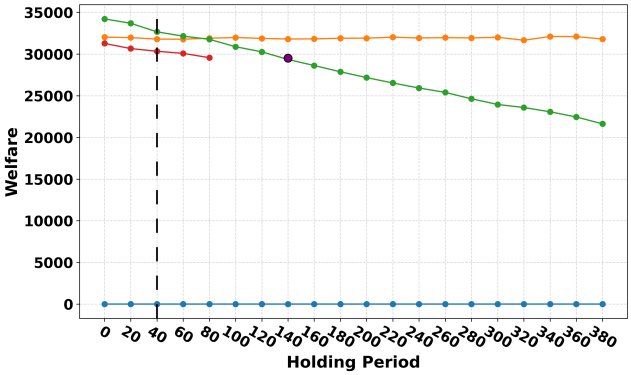


Figure 4: Total welfare in equilibrium: sum of the two plots in Figure 3.

(orange), MOBIs benefit from the M-ELO option (up to a high *HP* value) whereas background traders lose surplus.

These findings validate the central insight behind the M-ELO holding period, as it effectively deters the relatively fast traders from jumping into the dark pool. The practical challenge is to select a holding period long enough to discourage faster or more informed traders while maintaining acceptable delay costs for the traders for whom the mechanism is designed.

5.3 Adoption of M-ELO vs. Lit Market

We can examine the incentives at play in the market selection decision more closely by visualizing the relations among strategy profiles through ***deviation graphs***. Figures 5, 6, and 7 depict deviation graphs for the empirical DPR games at *HP* settings of 20, 120, and 380 respectively. Each cyan node represents a DPR profile, and each colored edge denotes the profitable single-player deviations for a role. Brown edges signify an improving deviation for a MOBI player, and blue edges the improving deviations for background traders.

5.3.1 Deviation Graph Holding Period 20. Two corner equilibria stand out. At (0, 4; 0, 4), all agents remain in the lit market because any deviation to M-ELO would isolate the deviator, making it unprofitable, a pattern observed across all holding periods. This equilibrium occurs consistently due to the isolation risk associated with deviating to an empty M-ELO. At (4, 0; 4, 0), all agents are in M-ELO here, a deviation to the lit market would similarly isolate the deviator. Moreover, in the absence of midpoint liquidity, all agents earn zero. As this profile offers no benefits to its neighbors, it is not an attractive equilibrium, and we consider it degenerate.

Interestingly, at this low holding period, there exists another pure equilibrium at (4, 0; 3, 1). In column 4, all blue background trader edges point upward, indicating deviations from the lit market to M-ELO. In this column all the MOBIs are in M-ELO, so background traders have opportunities to trade with them there. At $HP = 20$, the delay cost is not sufficient to deter them from entering the dark pool. In the other columns, almost all blue arrows point downward, suggesting that the concentration of MOBIs is necessary to attract the background traders here.

We can also find patterns along the rows. In rows 0, 1, 2, and 3 from column 1 onward, all brown arrows point rightward, indicating that MOBIs who are in the lit market profit by moving to M-ELO when that market is already populated with a MOBI trader. Column 1 shows the opposite pattern for a sole MOBI in M-ELO, who would prefer to be in the lit market when all the other MOBIs are there. Although they face adverse selection from smaller, more frequently arriving background traders in the lit market, this cost remains lower than trading in a weak M-ELO market. Row 4

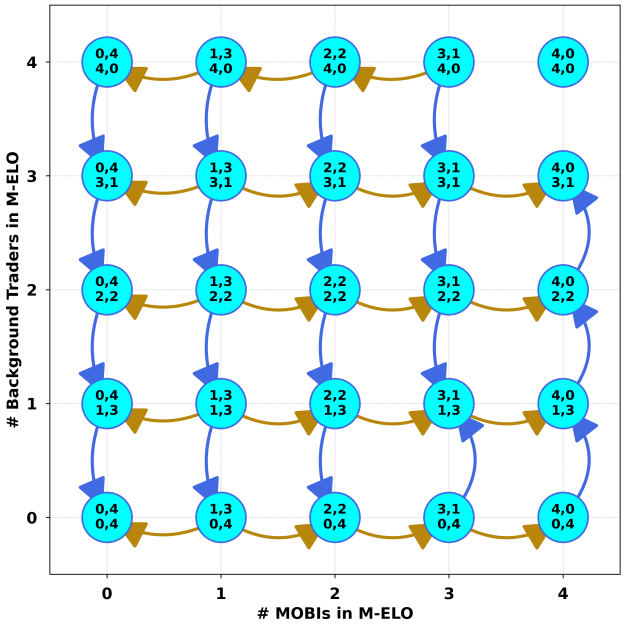


Figure 5: Deviation graph for the $(4+4)$-player DPR game at $HP = 20$. Each node is positioned at coordinates (n_1, n_2), the numbers of MOBI and background traders choosing M-ELO (with the remaining $4 - n_1$ and $4 - n_2$ choosing lit). Nodes are labeled with their strategy counts $(n_1, 4-n_1; n_2, 4-n_2)$. Colored arrows show profitable single-agent deviations: brown for a MOBI trader's deviations and blue for background traders'.

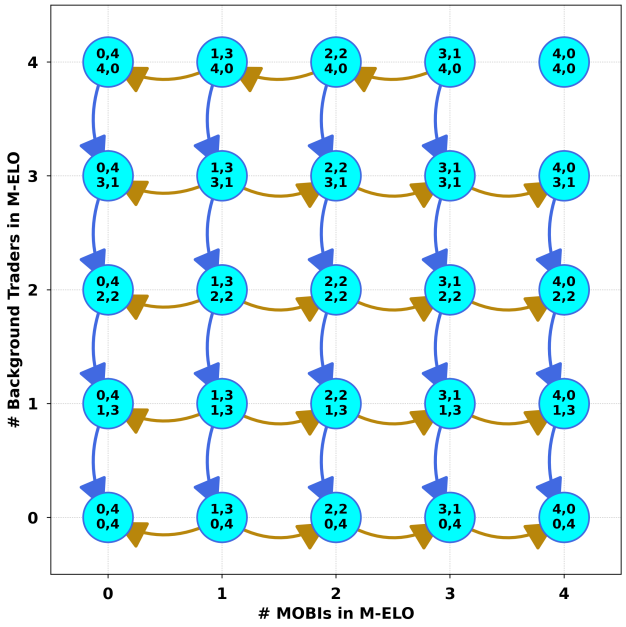


Figure 6: As Figure 5, but at $HP = 120$.

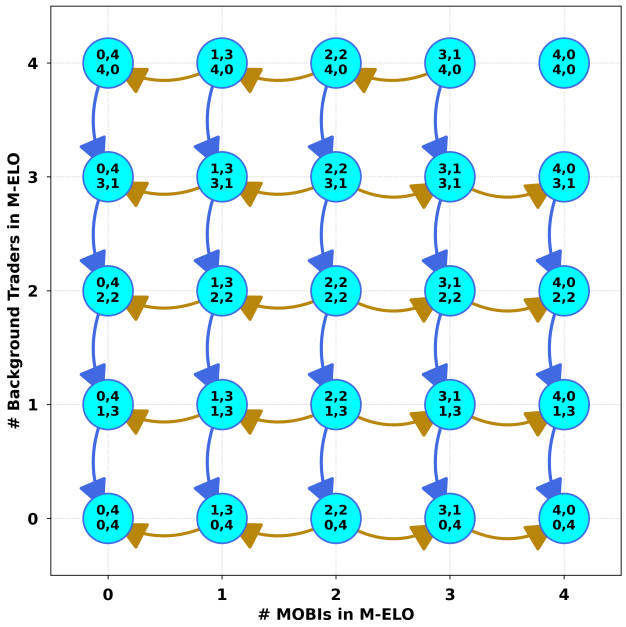


Figure 7: As Figure 5, but at $HP = 380$.

presents another (unlikely) scenario where the MOBIs would prefer the lit market: when all the background traders are in M-ELO, leaving the lit market open for MOBIs to trade without adverse selection risk. These patterns reflect M-ELO's design impact even at low holding periods. Population density emerges as the critical factor: insufficient population makes M-ELO more costly than facing adverse selection in the lit market, while adequate population transforms M-ELO into an effective refuge.

5.3.2 Deviation Graph Holding Period 120. Moving to a longer holding period of 120, many of the patterns seen at $HP = 20$ still hold, but key strategic shifts emerge. As always, the corner equilibria (0, 4; 0, 4) and (4, 0; 4, 0) equilibria remain. A shift can be seen in column 4. In $HP = 20$, all blue arrows in this column point upward, whereas in $HP = 120$ there is a complete reversal. This holding period is long enough to completely dissuade background traders from deviating to M-ELO, even when all the MOBIs are there. In fact, no blue arrows in this graph point upward; all best deviations are toward the lit market. Given this, we also have an equilibrium at (4, 0; 0, 4)—not present for $HP = 20$—with all MOBIs in M-ELO and all background traders in the lit market. The holding period is large enough to keep background traders out but still small enough to keep M-ELO worthwhile for MOBIs. Row-wise, there is no difference between holding periods 20 and 120. The lengthened holding period has not grown large enough to change the MOBIs' incentives; they still find M-ELO just as attractive at this higher holding period, but only if there are other MOBI traders already in M-ELO.

5.3.3 Deviation Graph Holding Period 380. At $HP = 380$ the deviation graph matches $HP = 120$ from the perspective of background traders. All blue arrows point downward, so they never gain by

switching to M-ELO. Not surprisingly, as $HP = 120$ deterred entry to M-ELO, $HP = 380$ would deter them all the more. The change is for MOBIs: in rows 0 to 3, column 2, the brown edge now points out, indicating that the longer wait deters them unless many MOBIs are already in M-ELO. When the holding cost outweighs adverse selection risk, traders of both roles abandon M-ELO entirely.

5.3.4 Basins of Attraction. These deviation graphs also facilitate understanding of the basins of attraction for each equilibrium. For every holding period the degenerate corner equilibrium (4,0;4,0), where all traders of both roles go to M-ELO, constitutes a one-profile basin. For the remaining 24 profiles, where their outward paths lead varies with *HP*.

For $HP = 20$, the pure lit equilibrium (0,4;0,4) pulls in every state with at most one MOBI in M-ELO. All other profiles, including every state that already has at least 2 DPR MOBIs in M-ELO, drift to the sink (4,0;3,1). Thus in the reduced-player game a single MOBI can tip most of the grid toward M-ELO. Raising the waiting time to $HP = 120$ leaves that lit basin intact but swaps the sink (4,0;3,1) for the sink (4,0;0,4): background traders now stay out of M-ELO, yet one MOBI is still enough to secure the M-ELO heavy outcome. By $HP = 380$ the balance flips: the corner where all 4 traders of each role go to the lit market now absorbs every profile with at least MOBIs already there (columns 0 to 2) plus the lone outlier (3,1;4,0), while the sink (4,0;0,4) governs all denser M-ELO states. In effect the critical mass of DPR players needed to anchor choice of M-ELO climbs from one at low holding period to three at very high holding period, showing that longer holding periods progressively erode M-ELO's basin unless a sufficiently large MOBI block coordinates there from the start.

6 Conclusion

We analyzed a trader's strategic decision to adopt a midpoint matching mechanism, considering tradeoffs in liquidity and adverse selection, under varying delay costs. Our model is based on Nasdaq's M-ELO mechanism, which we implemented alongside a traditional lit market in an agent-based financial market simulator. The model featured choices by two categories of traders, one relatively slow and high-volume (MOBI) and the other more nimble and low-volume (Background). We applied Empirical Game-Theoretic Analysis with Deviation Preserving Reduction to construct game models from simulation data over a range of mechanism settings (dark pool holding period) and exercising a variety of market choice combinations made by the agents. The analysis identified qualitatively distinct equilibria and characterized their appearance as a function of the holding period. Inspection of deviation-graph representations of our game models sheds light on agent-level incentives and the stability of alternative equilibria.

Overall, our study confirms intuitions underlying the design of midpoint matching mechanisms with holding periods. Sufficiently high holding periods effectively deter background traders from entering the dark pool, while still keeping that mechanism worthwhile for the MOBI traders. Shorter delays can fail to deter background traders, yet excessively long holding periods erode the midpoint premium for MOBIs. Even at intermediate settings, there are also equilibria where all traders stay in the lit market, indicating that a critical mass of MOBI adoption is necessary to sustain the dark mechanism.

This work highlights fundamental tradeoffs in holding period at the core of designing dark trading mechanisms. Further analysis and extension of these models may contribute to decision support for such mechanism design, and enable the development of mechanisms that ideally balance the efficiency and price discovery properties of financial markets.

Acknowledgments

We thank Drew Rooks for helpful discussions, guidance, and for answering our initial questions about M-ELO.

References

[1] Matteo Aquilina, Sean Foley, and Peter O'Neill. 2023. *Sharks in the Dark: Quantifying HFT Dark Pool Latency Arbitrage.* Technical Report 1115. BIS Working Papers. https://www.bis.org/publ/work1115.pdf

[2] Robert L. Axtell and J. Doyne Farmer. 2025. Agent-Based Modeling in Economics and Finance: Past, Present, and Future. *Journal of Economic Literature* 63, 1 (2025), 197–287.

[3] Anna Bayona, Ariadna Dumitrescu, and Carolina Manzano. 2023. Information and optimal trading strategies with dark pools. *Economic Modelling* 126 (2023), 106376.

[4] Michael Brolley. 2020. Price improvement and execution risk in lit and dark markets. *Management Science* 66, 2 (2020), 863–886.

[5] Sabrina Buti, Barbara Rindi, and Ingrid M. Werner. 2017. Dark pool trading strategies, market quality and welfare. *Journal of Financial Economics* 124, 2 (2017), 244–265.

[6] Sabrina Buti, Barbara Rindi, and Ingrid M. Werner. 2022. Diving into Dark Pools. *Financial Management* 51, 4 (2022), 961–994. doi:10.1111/fima.12395

[7] Carole Comerton-Forde and Tālis J. Putniņš. 2015. Dark Trading and Price Discovery. *Journal of Financial Economics* 118, 1 (2015), 70–92. doi:10.1016/j.jfineco.2015.06.013

[8] International Securities Exchange. 2006. *MidPoint Match To Be First Product.* Technical Report. U.S. Securities and Exchange Commission. https://www.sec.gov/Archives/edgar/data/1295230/000118143106024165/rrd114254_12996.htm.

[9] J. Doyne Farmer, Paolo Patelli, and Ilija I. Zovko. 2005. The predictive power of zero intelligence in financial markets. *Proceedings of the National Academy of Sciences* 102 (2005), 2254–2259.

[10] Dhananjay K. Gode and Shyam Sunder. 1993. Allocative Efficiency of Markets with Zero-Intelligence Traders: Market as a Partial Substitute for Individual Rationality. *Journal of Political Economy* 101, 1 (1993), 119–137.

[11] Jorge Gonçalves, Roman Kräussl, and Vladimir Levin. 2023. *Dark Trading and Financial Markets Stability.* Working Paper 691. Center for Financial Studies, Goethe University. Available at SSRN, 3384719.

[12] Anri Gu, Yongzhao Wang, Chris Mascioli, Mithun Chakraborty, Rahul Savani, Theodore L. Turocy, and Michael P. Wellman. 2024. The Effect of Liquidity on the Spoofability of Financial Markets. In *5th ACM International Conference on AI in Finance.* 239–247.

[13] H. Ishibuchi, R. Sakamoto, and T. Nakashima. 2001. Evolution of unplanned coordination in a market selection game. *IEEE Transactions on Evolutionary Computation* 5, 5 (2001), 524–534. doi:10.1109/4235.956715

[14] Diana Kafkes, Josep Puig Ruiz, Drew Rooks, Douglas Hamilton, and Michael O'Rourke. 2022. *Applying Artificial Intelligence & Reinforcement lLarning Methods Towards Improving Execution Outcomes.* Technical Report. Nasdaq Stock Exchange. Available at SSRN, 4243985.

[15] Michael Kaisers, Karl Tuyls, Frank Thuijsman, and Simon Parsons. 2008. Auction Analysis by Normal Form Game Approximation. In *2008 IEEE/WIC/ACM International Conference on Web Intelligence and Intelligent Agent Technology*, Vol. 2. IEEE, 447–450.

[16] Vladimir Levin. 2022. *Essays on Market Microstructure and Financial Markets Stability.* Ph. D. Dissertation. University of Luxembourg.

[17] Chris Mascioli, Anri Gu, Yongzhao Wang, Mithun Chakraborty, and Michael P. Wellman. 2024. A Financial Market Simulation Environment for Trading Agents Using Deep Reinforcement Learning. In *5th ACM International Conference on AI in Finance.* 117–125.

[18] Kevin S. Y. Mo, Mark Paddrik, and Steve Y. Yang. 2012. A Study of Dark-Pool Trading Using an Agent-Based Model. *SSRN Electronic Journal* (2012). doi:10.2139/ssrn.2181209

[19] Nasdaq, Inc. 2022. The Midpoint Extended Life Order (M-ELO) Frequently Asked Questions. https://www.nasdaqtrader.com/Content/ProductsServices/TRADING/M-ELO-FAQs.pdf. Accessed: 2025-07-15.

[20] Nasdaq Newsroom. 2023. Everything You Need to Know about Dynamic M-ELO, the First Exchange AI-powered Order Type. https://www.nasdaq.com/newsroom/dynamic-m-elo-ai-powered-order-type. [Online; accessed 19-July-2025].

[21] Nasdaq Trader. n.d.. M-ELO Transparency Statistics. https://www.nasdaqtrader.com/Trader.aspx?id=MELOSymbolData&utm_source. [Online; accessed 19-July-2025].

[22] New York Stock Exchange. 2009. *Routing to NYSE MatchPoint via the NYSE Common Customer Gateway.* Technical Report. New York Stock Exchange.

[23] L. Julian Schvartzman and Michael P. Wellman. 2009. Stronger CDA Strategies Through Empirical Game-Theoretic Analysis and Reinforcement Learning. In *8th International Conference on Autonomous Agents and Multiagent Systems*. 249–256.

[24] Megan Shearer, Gabriel Rauterberg, and Michael P. Wellman. 2023. Learning to Manipulate a Financial Benchmark. In *4th ACM International Conference on AI in Finance*. 592–600.

[25] Bing Shi, Enrico H. Gerding, Perukrishnen Vytelingum, and Nicholas R. Jennings. 2010. A Game-Theoretic Analysis of Market Selection Strategies for Competing Double Auction Marketplaces. In *9th International Joint Conference on Autonomous Agents and Multi-Agent Systems (AAMAS)* (Toronto). 857–864.

[26] Elaine Wah, Dylan Hurd, and Michael P. Wellman. 2016. Strategic Market Choice: Frequent Call Markets vs. Continuous Double Auctions for Fast and Slow Traders. *EAI Endorsed Transactions on Serious Games* 3, 10 (2016).

[27] Michael P. Wellman, Karl Tuyls, and Amy Greenwald. 2025. Empirical Game Theoretic Analysis: A Survey. *Journal of Artificial Intelligence Research* 82 (2025), 1017–1076.

[28] Bryce Wiedenbeck and Michael P. Wellman. 2012. Scaling Simulation-Based Game Analysis Through Deviation-Preserving Reduction. In *11th International Conference on Autonomous Agents and Multiagent Systems*. 931–938.

[29] Mason Wright and Michael P. Wellman. 2018. Evaluating the stability of non-adaptive trading in continuous double auctions. In *17th International Joint Conference on Autonomous Agents and Multi-Agent Systems (AAMAS)* (Stockholm). 614–622.

[30] Linlin Ye. 2016. Understanding the Impacts of Dark Pools on Price Discovery. arxiv:1612.08486. https://arxiv.org/abs/1612.08486

[31] Haoxiang Zhu. 2014. Do Dark Pools Harm Price Discovery? *Review of Financial Studies* 27, 3 (2014), 747–789. doi:10.1093/rfs/hht078